

Building Detection from Satellite Imagery using YOLOv8

1. Problem Statement

The objective of this project is to automatically detect and count buildings in satellite imagery using the YOLOv8 object detection model. Accurate building detection can support urban planning, disaster management, infrastructure monitoring, and population estimation. The final goal is to build a reusable pipeline that takes a satellite image as input and returns the detected buildings along with their count.

2. Dataset

The SpaceNet Buildings YOLO Dataset was used for this project. The dataset contains satellite image patches and corresponding annotation files in YOLO format. Each annotation file stores bounding box coordinates representing building locations within the image.

Dataset Statistics

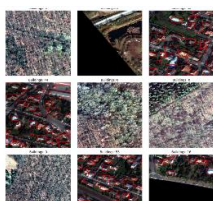
- Total Images: 1148
- Total Label Files: 1148
- Total Buildings: 16,633
- Average Buildings per Image: 14.49

The dataset was divided into:

Split	Number of Images
Train	918
Validation	115
Test	115

Dataset Observations

Most satellite images have consistent resolution and quality, which helps the model learn common building patterns. However, some challenges include densely packed buildings, small rooftops, shadows, and buildings with irregular shapes that may be difficult to distinguish from surrounding structures.



3. Model

YOLOv8s (You Only Look Once Version 8 - Small) was selected for this project because it provides a good balance between accuracy and computational efficiency.

Instead of training from random initialization, pretrained weights (yolov8s.pt) were used. Pretrained models have already learned useful visual features from large datasets, enabling faster convergence and better performance on the target task.

Model Configuration

- Model: YOLOv8s
 - Classes: 1 (Building)
 - Framework: Ultralytics YOLOv8
 - Approximate Parameters: 11 Million
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4. Training

The model was trained using Google Colab with a Tesla T4 GPU.

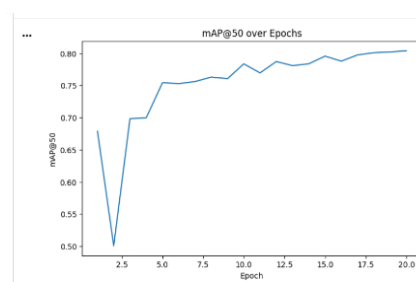
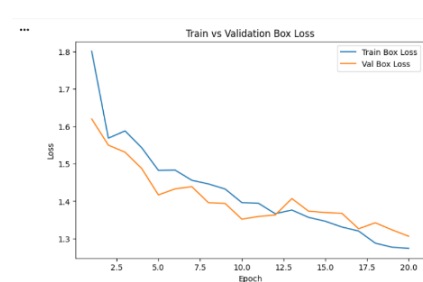
Training Configuration

- Epochs: 20
- Batch Size: 16
- Image Size: 640 × 640
- GPU: Tesla T4
- Optimizer: Default Ultralytics Settings

Training Results

- Final Box Loss: 1.27379
- Final mAP@50: 0.80376

The training loss decreased steadily over epochs, while the mAP@50 score improved and eventually began to plateau, indicating that the model was approaching convergence.



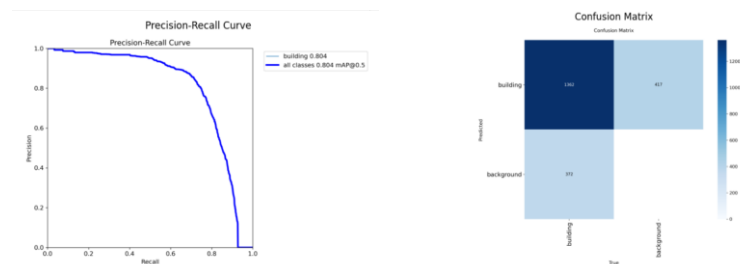
5. Evaluation

The trained model was evaluated on the held-out test set.

Performance Metrics

Metric	Value	Meaning
Precision	0.8544	Percentage of predicted buildings that were correct
Recall	0.7595	Percentage of actual buildings successfully detected
F1 Score	0.8042	Balance between precision and recall
mAP@50	0.8360	Overall object detection accuracy

The model achieved strong detection performance, with high precision and a good balance between precision and recall.



6. Building Detection Pipeline

A reusable building detection function was developed using the trained YOLOv8 model.

The pipeline performs the following steps:

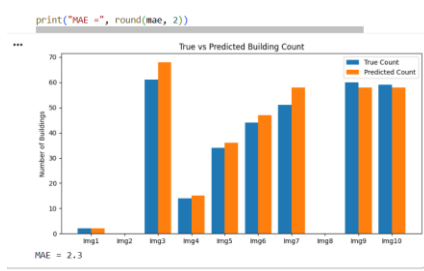
1. Accepts a satellite image as input.
2. Runs building detection using the trained model.
3. Draws bounding boxes around detected buildings.
4. Counts the total number of detected buildings.
5. Calculates average confidence scores.
6. Displays the annotated image and summary statistics.

Counting Accuracy

To evaluate counting performance, predictions were compared with ground-truth building counts on 10 test images.

- Mean Absolute Error (MAE): 2.3

This indicates that the predicted building count differed from the true count by approximately 2 buildings per image on average.



Sample Detection Result

The final pipeline was tested on multiple unseen test images and successfully detected buildings while reporting count, confidence score, and processing time.

7. Observations and Future Improvements

The model performed well on medium-sized and clearly visible buildings. It achieved an mAP@50 of 0.8360 and demonstrated strong precision and recall on the test set.

However, performance was slightly reduced in cases involving dense building clusters, small rooftops, and visually complex scenes. Future improvements could include training for more epochs, applying stronger data augmentation techniques, and experimenting with larger YOLOv8 variants such as YOLOv8m or YOLOv8l.

Overall, the project successfully achieved its objective of building an end-to-end building detection and counting system using satellite imagery and YOLOv8.